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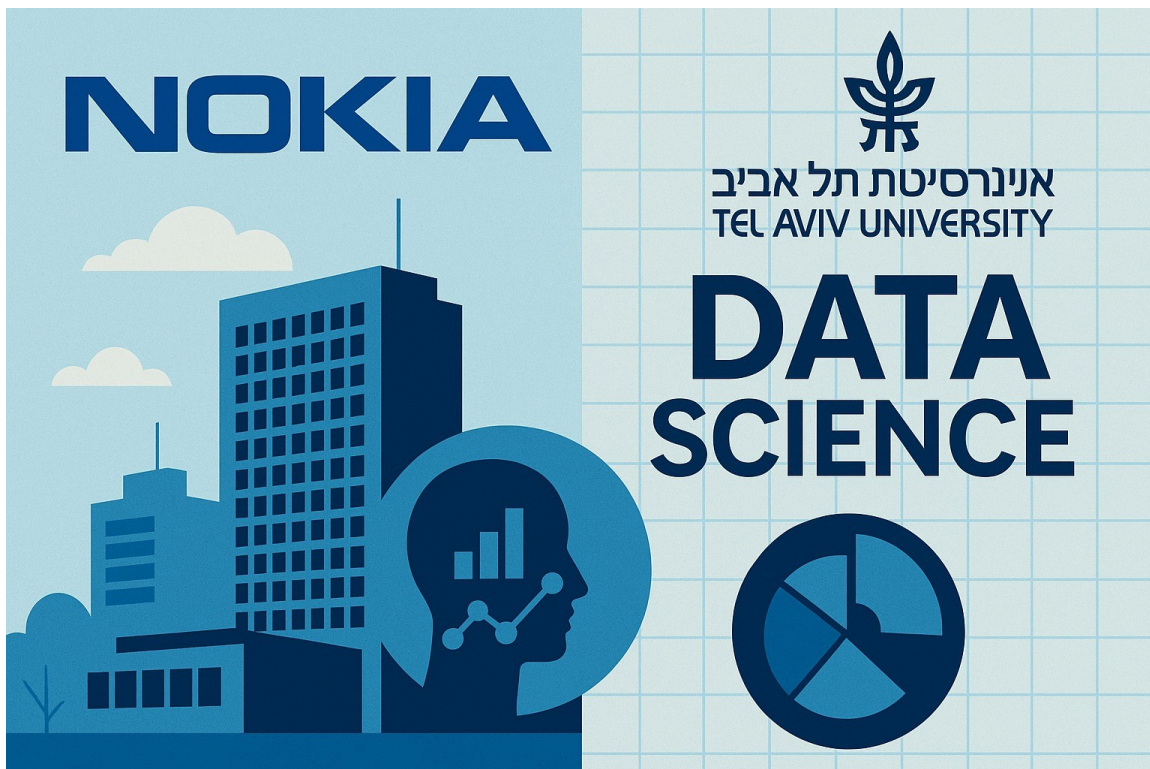
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NOKIA

Final Project Report



2024/2025

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Industry Partner: Liat Eilat Shalev,



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Project Title: **Predicting the Responsible Team & Groups for
Software Fault Resolution**

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Institution: Tel-Aviv University

Advisor: Tanya Zhelezniak

Submission Date:



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1. Industry Background

Overview:

Nokia operates within the telecommunications industry, specifically focusing on the development and provision of advanced communication infrastructure. The telecommunications sector is a key component in enabling global connectivity through cellular, internet, and emerging technologies, impacting billions of users worldwide.

Company ID¹:

- Established in 1865
- Global Presence: Employs around 80,000 people worldwide, with approximately 150 employees in Israel.
- Financial Highlights: Achieved a net profit of €4.3 billion in 2022.
- Market Scope: Supports of about 6.6 billion cellular users globally.
- Business Model: Primarily operates as a B2B (Business-to-Business) provider, offering infrastructure solutions directly to other companies rather than individual consumers.
- Technological Innovation: Involved in cutting-edge technology areas including IoT (Internet of Things), drones, autonomous vehicles, and advanced communication systems.
- Notable Achievements: Affiliated with prestigious entities like Siemens and Bell Labs and credited with contributing to the awarding of 9 Nobel Prizes.

Why is the topic important?

Efficient fault resolution is a critical challenge in any large-scale technological environment, especially within industries that rely on complex infrastructure and global support operations. As the volume and complexity of service tickets continue to grow, manual triage

¹The information is from Nokia Company Profile Presentation.



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becomes increasingly time-consuming, inconsistent, and costly.

Predictive models that can automatically suggest the most appropriate team or department to handle a given issue have the potential to streamline operations, reduce downtime, and enhance customer satisfaction.

Beyond operational benefits, the economic impact is significant: faster ticket resolution directly translates into reduced support costs, increased developer productivity, and improved resource allocation. In large organizations, even small improvements in mean time to resolution (MTTR) can result in substantial annual savings. This type of automation is not only relevant to Nokia but can be applied broadly across industries such as IT services, telecommunications, cloud computing, and enterprise support—where ticket handling is a major operational and financial concern.

Similar problems have been addressed in previous research, particularly in the fields of automated ticket routing and bug triage using machine learning. Numerous studies have explored ways to optimize the assignment of software faults or IT support tickets to the most appropriate team or engineer. For instance, the **TaDaa system** [1] employs Transformer-based deep learning models to assign tickets and suggest similar past issues, enhancing resolution efficiency. Other research has investigated the use of natural language processing and ensemble methods to improve fault assignment based on textual descriptions [2]. Additionally, hybrid systems combining machine learning with rule-based engines have been proposed for intelligent ticket handling in service environments [3]. These studies underscore the broader relevance of our work and reflect a growing interest in leveraging data-driven solutions for scalable operational improvements.



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2. The business problem

The business problem we have been trying to solve in this project is what team is needed to solve and close the ticket at Nokia, which significantly impacts the time a ticket stays open and impacts the operational efficiency and customer satisfaction.

Currently, Nokia faces approximately 10 new issues daily, with an average resolution period extending up to 30 days per issue. This extended resolution timeline leads to several critical challenges:

- **Excessive developer workload:** Valuable development time is disproportionately consumed by ongoing issue resolution, reducing productivity in core areas.
- **Workflow disruptions:** Persistent issues and delays frequently interrupt operational workflows, creating inefficiencies across teams.
- **Customer impact:** Issues that are unresolved or inadequately addressed sometimes reach customers, negatively affecting customer experience and potentially harming Nokia's reputation.
- **Limited scalability:** The inefficient resolution processes constrain scalability across various departments, limiting overall organizational growth and agility. Additionally, there are workers at Nokia dedicated solely to understanding the tickets and determining to whom they should be assigned.

The goal of this project is to analyze historical ticketing and changelog data to proactively predict the necessary team for ticket resolution from the start. By accurately forecasting the appropriate team assignments, Nokia aims to substantially reduce resolution times,



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enhance internal workflows, improve customer satisfaction, and facilitate better scalability across the organization.

3. Research Question

Initial question: Can obstacles in the fault resolution process be predicted?

Final question: Can machine learning optimize issue resolution times in telecommunications?

The core objective of this project was to improve the fault resolution process by predicting the team or group responsible for handling each issue. To guide our approach, we formulated several measurable research directions:

1. Predicting potential obstacles in the resolution process, such as delays, misassignments, or excessive handovers between teams, based on historical patterns.
2. Developing a model capable of identifying the responsible handling team or group at the time a ticket is created, using only the information available in the early stages.
3. Analyzing which features - both structured (e.g., components, network elements) and unstructured (e.g., textual descriptions) - have the greatest impact on the prediction of team assignment.
4. Evaluating the predictive power of textual descriptions alone by transforming them into vector embeddings and assessing their standalone contribution to model performance.
5. Comparing the performance of team-level versus group-level classification to determine whether aggregating teams into



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broader categories leads to more accurate and generalizable results.

These research directions served as the foundation for our data exploration, modeling decisions, and performance evaluations throughout the project.

4. Data Sources and Collection²

The data used in this project was provided by the company through its Jira system and includes three main datasets:

Changelog Dataset:

Tracks the history of ticket updates, capturing every change made to a ticket.

Key Data Elements:

- o ID: Unique identifier for each change record
- o Issue_Key: Unique identifier for the ticket
- o FIELD: The attribute that was changed (e.g., status, team)
- o OLDSTRING / NEWSTRING: The previous and updated values of the field
- o CREATED: Timestamp of when the change occurred

Core Dataset:

Contains static information about each ticket.

Key Data Elements:

- o Key: Unique identifier for each ticket
- o Severity: Issue criticality (Critical, Major, Minor)

² Based on Nokia Jira changelog, ticket statuses, and comments spreadsheet.



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- o description: Summary of the issue resolution
- o Root Cause Type and Root Cause Type.child: Main and subcategories of the issue's root cause
- o Components, Sub-Components, and Network Element: Technical entities involved in the issue

Comments Dataset:

Captures the textual comments associated with each ticket, reflecting team discussions and updates.

Key Data Elements:

- o Key: Unique identifier for each ticket
- o Total Comments: Total number of comments recorded
- o Comment1, Comment2, ... CommentN: Individual comments related to the ticket

These datasets collectively provide a comprehensive view of the issue lifecycle, from creation to resolution, enabling analysis of workflows, delays, and team interactions.

5. Exploratory Data Analysis (EDA)

Our primary focus during the first semester was on thoroughly exploring and understanding the data. We aimed to extract insights and highlight the most important and business-critical aspects like issues categorized as Critical or Major in severity or how many tickets were transferred between multiple teams (more than 2 teams) (Fig. 1).

We also leveraged the Changelog Dataset to derive operational insights and use cases that informed our exploration and modeling. Specifically, we:



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- Analyzed team handoff frequency to optimize workflow allocation
- Identified bottlenecks based on repeated status patterns
- Prioritized high-severity issues for deeper investigation
- Measured time in status to pinpoint delays and improve efficiency

This phase included:

- **Issue Lifecycle Analysis:** Investigating how tickets transition between different statuses and the duration they spend in each.
- **Severity-Based Trends:** Comparing resolution times across issues marked as Critical, Major, or Minor to understand priority-based behavior.
- **Team Assignment Patterns:** Evaluating the frequency and impact of team reassignments on issue resolution times.
- **Status Transition Counts:** Measuring how often each status appeared to identify frequent blockers or inefficiencies.

These exploratory insights were crucial for identifying bottlenecks, selecting features for modeling, and shaping strategies for predicting the correct team assignment from the start.

6. Data Cleaning and Preparation

To ensure the data was ready for modeling, we applied several preprocessing steps to the Main Dataset, which was the only dataset used for modeling:

- **Handling Missing Values:** Removed entries with missing or invalid fields such as team assignment, timestamps, or field change values.



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- **Filtering Non-Fault Tickets:** We excluded tickets that were not labeled as faults (e.g., service requests), using the field `fields.Item Type.value`. This ensured that the model was trained only on software-related incidents, which are the true focus of this classification task.
- **Filtering Open or Unresolved Tickets:** Tickets that were still open or had not yet reached a final resolution were filtered out using `fields.status.name`. Since the final responsible team is not yet known for these cases, they cannot be reliably used for supervised learning.
- **Feature Engineering:** Created new variables, including:
 - Number of team changes per ticket
 - Total number of status changes
 - Time spent in each status
 - Flags for transitions involving critical statuses (e.g., Open → In Progress)
- **Status Merging and Filtering:** With guidance from Nokia, we merged and removed status values that were rarely used, redundant, or not considered relevant to the company's main operational focus for this project.
- **Removing Post-Closure Updates:** Features that were only filled after the ticket was closed or retroactively updated were excluded to prevent data leakage and ensure real-time applicability.
- **Outlier Filtering:** Identified and removed tickets with extreme or inconsistent status patterns that could bias the model.



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- **Categorical Encoding:** Encoded relevant fields such as FIELD, OLDSTRING, and NEWSTRING to allow inclusion in the model pipeline.

Following the EDA phase and the insights derived from our initial analysis, we collaborated with Nokia to clearly define the core problem and align on the next step: **modeling the issue more accurately.**

Through this collaboration, we refined the problem definition and clarified the modeling objective - to develop a model capable of accurately predicting the appropriate team assignment for each ticket based on its attributes.

As a result, we received an updated dataset from Nokia, better aligned with the conclusions drawn from the initial EDA phase.

In addition, we applied several further preprocessing steps to the new dataset:

- **Feature Simplification and Real-Time Input Selection:** As part of reducing the complexity of the input data and improving model generalizability, we conducted a manual feature selection process in collaboration with a domain expert from Nokia. The goal was to retain only fields that are reliably and consistently filled out at the moment a new issue is opened—ensuring that our model would be applicable in real-world, early-stage scenarios. We selected a small set of key features, including fields.description, fields.components, fields.Original Network Element/s, and fields.Team.name, which are known to be present and trustworthy during the initial ticket submission. In addition, we manually mapped specific teams into groups to simplify the classification space. This streamlined approach ensured that the



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model focused only on actionable, real-time information available at the moment of fault creation.

- **Dropped features with low relevance or redundancy** - such as issue descriptions, sub-components, and item types-based on domain knowledge and with guidance from Nokia.
- **Applied one-hot encoding** to categorical features such as components and original network elements to convert them into a model-compatible numerical format.
- **Performed label encoding** on the target variable (team ID) to convert team identifiers into integers suitable for classification tasks.
- **Embedding Integration:** As a first step in the modeling pipeline, we transformed the **textual issue descriptions** into dense vector representations using **OpenAI's GPT-3 embedding model** (text-embedding-ada-002). These embeddings captured semantic context and were **concatenated with structured features**, forming a unified input vector for the model.
- **Data Loaders:** The combined data (structured features + GPT embeddings) was converted into **PyTorch tensors** and wrapped in DataLoader objects to support batching, shuffling, and GPU acceleration during training and evaluation.
- **Group Mapping, Data Filtering, and Final Feature Assembly:** To further simplify the classification space and improve model performance, we introduced an intermediate step that mapped specific fields.Team.id values into broader team groups based on an external CSV file (for the students.csv). This mapping was done using a dictionary (team_to_group) that grouped teams with shared characteristics or responsibilities.



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Following this, we: Filtered out ungrouped samples by dropping rows where no group mapping was available. Retained only groups with sufficient support (at least 10 samples) to ensure statistical validity during training and evaluation. Removed rare teams (teams with fewer than 5 occurrences) to avoid overfitting to noise and improve generalizability. Next, we constructed the final input feature matrix by: Stacking the GPT-3 embedding vectors extracted from the issue descriptions. Applying one-hot encoding to categorical fields such as fields.components and fields.Original Network Element/s, replacing missing values with a placeholder. Concatenating the embedding matrix with the one-hot encoded features to create a unified numerical feature space. The final label vector y was created by mapping the team IDs to their respective groups and applying label encoding to transform group names into integer values suitable for classification.

- We used a **train-validation split** strategy to achieve **more stable results**, due to the sparsity and class imbalance observed during earlier cross-validation experiments.

7. Definitions of Concepts and Parameters

This section provides concise definitions of key machine learning concepts and parameters used in this project.

- **Neural Network (NN):** A type of machine learning model inspired by the human brain, consisting of layers of interconnected nodes (neurons) used to capture complex patterns in data.
- **Embedding:** A numerical vector representation of textual or categorical data that preserves semantic meaning and enables input into machine learning models.



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- **Learning Rate:** A hyperparameter that controls how much the model weights are updated during training. Too high may cause divergence, while too low may slow convergence.
- **Batch Size:** The number of training samples used in one forward/backward pass. Affects training speed and model generalization.
- **Weight Decay:** A regularization technique that penalizes large weights to help prevent overfitting and improve model generalization.
- **Hyperparameters:** Configuration values set before training that affect the learning process (e.g., number of layers, learning rate). These are tuned to optimize performance.
- **Hyperparameter Tuning (Optuna):** An automated method to search for optimal hyperparameter values using intelligent sampling strategies.
- **Cross-Entropy Loss:** A loss function commonly used in classification tasks, measuring the difference between predicted probabilities and true class labels.
- **Adam Optimizer:** An optimization algorithm that combines momentum and adaptive learning rates for efficient and robust gradient descent.
- **Early Stopping:** A regularization technique that halts training when performance on validation data stops improving, to prevent overfitting.
- **Train/Validation Split:** Dividing the dataset into separate parts for model training and evaluation, ensuring performance is measured on unseen data.



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- **AUC-ROC:** Area Under the Receiver Operating Characteristic Curve – a metric that evaluates the model's ability to distinguish between classes.
- **Precision, Recall, F1 Score:** Metrics that assess the quality of classification, particularly in imbalanced datasets.
 - $Precision = TP / (TP + FP)$
 - $Recall = TP / (TP + FN)$
 - $F1\ Score = \text{Harmonic mean of precision and recall.}$

8. Initial Attempts and Lessons Learned

Our initial approach involved training a simple neural network with a single hidden layer to predict the responsible team for each issue, out of a total of 68 unique teams. The model achieved a validation accuracy of **42.8%**, with a low F1 score of **0.18**, indicating limited ability to correctly classify across such a large number of classes.

We then increased the network complexity by adding multiple hidden layers and tuning the number of layers (n_layers) as a hyperparameter. However, the results remained suboptimal, with average accuracy dropping slightly to **40.4%** and the F1 score to **0.16**, suggesting overfitting and instability due to class sparsity.

To address this, we redefined the prediction target: instead of predicting the exact team, we grouped teams into broader categories ("groups") and trained the model to predict the appropriate group. This change simplified the classification task and significantly improved performance — accuracy rose to **61.2%**, and the F1 score increased to **0.59**.



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Finally, we refined our hyperparameter ranges based on performance insights from W&B logging. After tuning, the best-performing model (group-level prediction with optimized hyperparameters) achieved a validation accuracy of **61.2%**, F1 score of **0.59**, and an AUC-ROC of **0.905**, indicating a robust and generalizable solution.

9. Model Training and Selection Process

Model Architecture:

We implemented a custom **feedforward neural network**, TabularNN, using PyTorch. The architecture consisted of:

- **Input Layer:** Accepts the concatenated feature and embedding vector
- **Hidden Layers:** A variable number (n_{layers}) of fully connected layers, each with:
 - Linear transformation
 - ReLU activation
- **Output Layer:** One neuron per class (either team or group), producing logits for classification

Layers were stacked dynamically using `nn.Sequential`, allowing us to explore various depths while maintaining flexibility and interpretability.

Hyperparameter Tuning (Fig. 2.1)

To optimize model performance, we used **Optuna** for automated hyperparameter tuning, guided by performance insights from Weights & Biases (W&B).



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Initial Search Space : We began with a broad search space informed by prior knowledge and default ranges commonly used in deep learning tasks. The initial ranges were:

- **hidden_dim:** Number of neurons per hidden layer (64–768)
- **n_layers:** Number of hidden layers (2–4)
- **learning_rate:** Learning rate (1e-5 to 5e-4, log-uniform)
- **batch_size:** Number of samples per batch (32–128)
- **weight_decay:** L2 regularization coefficient
- **patience:** Number of non-improving epochs before triggering early stopping (3–10)

Observations and Adjustments

After analyzing the performance distribution of trials via **W&B visualization tools** (such as parallel coordinate plots and performance histograms), we identified regions in the hyperparameter space that were consistently underperforming or had minimal impact on the objective metric.

Refined Search Space : We then narrowed the search space to focus on the most promising ranges, improving tuning efficiency and result stability:

- **hidden_dim:** 256–640 (step of 64) — focused on mid-high capacities that showed consistent gains
- **n_layers:** 3–4 — deeper models demonstrated better generalization
- **learning_rate:** 1e-4 to 5e-4 — excluded extremely small values that slowed learning



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- **batch_size**: 64-128 — larger batches led to more stable convergence
- **weight_decay**: $2e-5$ to $5e-5$ — focused on regularization values that reduced overfitting without underfitting
- **patience**: 5-8 — tuned based on early stopping behavior observed in training curves

This two-phase tuning approach allowed us to balance **exploration** and **exploitation**, improving both **model robustness** and **training efficiency**.

Training Procedure

For each trial:

- **Initialization**: A new TabularNN model was built using the sampled hyperparameters.
- **Optimizer**: **Adam**, chosen for its adaptive learning capabilities.
- **Loss Function**: **CrossEntropyLoss**, appropriate for multiclass classification.
(Fig 2.2, Fig 2.4)
- **Metric Tracking**: Throughout training, we logged metrics using **Weights & Biases (W&B)**:
 - Accuracy (Fig 2.3, Fig 2.5)
 - Precision (Fig 2.7)
 - Recall (Fig 2.6)
 - F1 Score (macro) (Fig 2.8)
 - AUC-ROC (for group-level predictions)



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Training proceeded over multiple epochs, with performance monitored on the validation set. **Early stopping** was applied based on validation loss to prevent overfitting and reduce computational overhead.

Model Selection and Hyperparameter Optimization (Fig. 3)

This section presents the results of model training and hyperparameter optimization, evaluating performance across multiple trials using key metrics such as accuracy, loss, precision, recall, F1 score, and AUC-ROC. The goal was to identify the most effective configuration for predicting the responsible group for software fault resolution. Visualizations generated through Weights & Biases (W&B) allow us to assess convergence behavior, training dynamics, and the impact of hyperparameters on performance.

The **Validation Loss** graph shows that nearly all trials demonstrate a consistent decline in loss during training, indicating effective learning and convergence. Most models stabilize around a loss value between 1.1 and 1.2 after approximately 10 steps. A few trials (e.g., trial_23) started with higher loss and converged more slowly, but eventually aligned with the general trend. The clustering of lines in the later stages suggests that multiple hyperparameter configurations produced similarly good generalization performance.

The **Validation Accuracy** graph shows consistent upward trends across nearly all trials, with most models converging to an accuracy range between 0.58 and 0.62. The early epochs demonstrate rapid learning, followed by a plateau as models stabilize. The close clustering of final accuracy values indicates that multiple hyperparameter configurations led to similarly strong generalization performance, with no extreme outliers or major underperformers.

The **Train Loss** graph shows a consistent and steady decline across nearly all trials, indicating successful learning on the training data. Most models reduced loss significantly within the first 5–10 epochs, with many reaching values below 1.0 by the end of training. This uniform downward trend reflects effective convergence, and the tight clustering of curves suggests that multiple hyperparameter configurations were able to fit the training set well without severe overfitting or divergence.



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The **Train Accuracy** graph shows a steady upward trend across most trials, with several models reaching training accuracy levels above 70% by the final epoch. This consistent improvement reflects effective learning and convergence. While some variation exists between trials, particularly in early epochs, the majority show clear progress, indicating that the training configurations were generally successful in fitting the data.

The **Recall** graph shows a consistent upward trend across nearly all trials, with most models reaching recall values between 0.5 and 0.6 by the final epoch. This indicates improved ability to correctly identify relevant classes over time. While some variation remains, the overall convergence suggests that the models are learning to better capture true positives during training.

The **Precision** graph reveals a noticeable drop during early epochs across most trials, stabilizing later between 0.65 and 0.75. This initial decline suggests that as models learned to identify more positive samples (increased recall), they temporarily sacrificed precision. Over time, however, precision recovered and converged across trials, reflecting improved balance in class predictions and consistent model refinement.

The **F1 Score** chart shows steady and consistent improvement across most trials, reflecting an increasingly balanced trade-off between precision and recall. While early epochs display wide variability, the majority of trials converge to F1 scores around 0.55–0.60 by the end, indicating solid overall performance. This suggests that the models not only captured relevant classes but also maintained precision in doing so, leading to well-rounded predictive capabilities.

Comparing the initial and final parallel coordinates plots, we observe a clear evolution in hyperparameter optimization strategy. In the first tuning phase, high-performing trials (yellow lines indicating low validation loss) were scattered and less concentrated, with broader variation across parameters such as `batch_size`, `hidden_dim`, and `learning_rate`. Notably, optimal results were harder to isolate, reflecting early-stage experimentation with diverse configurations.

In contrast, the final tuning phase shows much tighter clustering of high AUC-ROC scores (also marked in yellow) around specific hyperparameter ranges—particularly `batch_size` of 120–130,



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hidden_dim of 500–600, learning_rate of ~ 0.00035 – 0.0005 , and $n_layers = 3$. This convergence indicates that the optimization process successfully narrowed down effective configurations, leading to more stable, generalizable model performance.

In conclusion, the hyperparameter optimization and training process yielded a robust and reliable model for predicting the responsible group in software fault resolution. The convergence observed across metrics such as loss, accuracy, precision, recall, and F1 score confirms the effectiveness of our modeling approach and the consistency of performance across diverse configurations. The refinement in hyperparameter selection over time, as evidenced by the parallel coordinates plots, further demonstrates that targeted tuning can significantly improve model generalization. These results validate the model's potential for practical deployment and set a strong foundation for future iterations and enhancements.

10. Model Results and Performance

To evaluate the model's learning process and ensure it generalized well after hyperparameter tuning (Fig 4.1), we retrained the final model on the full training and validation set. The following metrics were tracked throughout the training:

accuracy (Fig 4.3), F1 score (Fig 4.4), precision (Fig 4.2), and loss (Fig 4.5). These metrics offer a comprehensive view of how well the model fit the data, both in terms of raw correctness (accuracy), robustness across class imbalance (F1), reliability of predictions (precision), and convergence of the optimization process (loss).

The training loss decreased steadily and significantly over the course of training, starting above 1.4 and reaching close to zero by the final epoch. This consistent downward trend indicates that the model successfully minimized its error on the combined training and validation data. The smoothness of the curve, with no sudden jumps or oscillations, also suggests stable optimization and convergence.

The final training precision curve shows a clear upward trend, starting around 58% and reaching approximately 94% by the end of training. Despite some minor fluctuations in early epochs, the overall increase suggests that the model became progressively more effective at minimizing false positives. The high final precision score indicates that



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the model confidently assigns correct labels to the predicted classes, further reinforcing the reliability of its decisions on the training data.

The chart illustrates the training accuracy of the final model over successive epochs. The model demonstrates a steady and consistent improvement in accuracy, starting at around 47% and reaching approximately 95% by the final training step. This sharp upward trend suggests that the model successfully learned from the combined training and validation data, with no signs of underfitting. However, the high final accuracy may also indicate potential overfitting, which should be evaluated against test set performance.

The F1 Score progression during the final training phase shows a continuous improvement, rising from approximately 30% to nearly 90% by the end of training. This indicates that the model not only improved in overall accuracy, but also maintained a strong balance between precision and recall across classes. The F1 score curve stabilizes in the later epochs, suggesting convergence. These results reflect robust performance on the combined training and validation set, though generalization should be confirmed via test set metrics.

Summary and Interpretation

The training results demonstrate strong and stable model learning. Accuracy and F1 score steadily improved and converged at high levels (around 95% and 90%, respectively), indicating that the model captured both correct and balanced predictions across classes. Precision rose consistently, suggesting a reduction in false positives, while the training loss decreased smoothly, confirming effective and stable optimization. Together, these metrics reflect a well-performing model with low training error and reliable classification behavior, setting a strong foundation for final evaluation on the unseen test set.

Confusion Matrix - Test Set (Fig. 5)

The confusion matrix provides a detailed view of the model's classification behavior on the test set. Most classes show a strong concentration of predictions along the diagonal, indicating correct predictions. Notably, groups like **CNF**, **Charging**, **UI**, and **SS7** demonstrate high true positive counts, reflecting strong model confidence and accuracy in those categories. However, there are some misclassifications between similar or related groups—for example, several **Charging** samples were predicted as **CCoE** or **Support**, which



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may suggest overlapping features or label ambiguity. These insights highlight both the model's strengths and specific areas for potential refinement.

Group-wise Performance Metrics (Fig. 6)

The performance bar chart breaks down key evaluation metrics—precision, recall, F1-score, and accuracy—for each group. The model achieved excellent scores on **UI**, **SS7**, and **CNF**, where all metrics exceeded 0.85, indicating both precise and comprehensive classification. In contrast, groups like **SPV**, **Support**, and **DevOps** showed lower and less balanced scores, especially in recall and F1, likely due to limited support in the test set or overlapping patterns with other teams. These group-level insights are valuable for guiding future data balancing or targeted model improvements.

Overall Test Set Metrics

On the test set, the model achieved an overall **accuracy of 61.8%**. The **macro-averaged precision** was **0.6387**, with a **macro recall** of **0.6012** and **macro F1-score** of **0.6102**, indicating reasonably balanced performance across all classes, regardless of their size. The **weighted averages**, which account for class imbalance, were slightly lower and more aligned with accuracy:

- **Precision (weighted avg):** 0.6167
- **Recall (weighted avg):** 0.6179
- **F1-score (weighted avg):** 0.6150

These values suggest that while the model performs fairly well on dominant classes, additional improvements could be made to boost generalization across minority groups.

To qualitatively illustrate the model's performance on the test set, we present a table containing up to five examples from each category of the confusion matrix: True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN). These examples highlight cases where the model classified incidents correctly or incorrectly,



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providing deeper insights into its strengths and weaknesses across different scenarios.

Example Type	Component	Original NE	Description (Truncated)	Predicted Group	Actual Group
True Positive (TP)	CNF	None	Working Directory: /root/CNF/WORK/... HTTP ERROR 500 ... NoSuchBeanException ...	DR	DR
False Positive (FP)	RRM	None	Type of Issue: < ME / EME / RCA > ... Steps to Reproduce: ...	DR	Support
False Negative (FN)	Charging	None	We are not getting SNR in the pcap traces ... FASTPASS5GB bundle ...	Charging	DR
True Negative (TN)	CNF	None	indexsearch-secadmin pods are not up ... configmaps "indexsearch-osinstalled" not found	CNF	CNF

Trade-offs

Throughout the project, we made strategic tradeoffs to balance performance, fairness, and real-world applicability. First, we predicted at the group level rather than the team level. This reduced prediction granularity but improved accuracy from ~42% to over 60% by addressing label sparsity and class imbalance. Second, we favored model performance over interpretability by using a neural network with GPT-based embeddings. While this introduced a black-box element, it



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enabled the model to capture complex patterns in both structured and unstructured data, boosting F1 and AUC-ROC scores.

To allow for Practical implementation, we limited features to those available at ticket creation, avoiding reliance on post-resolution data. This may have reduced potential accuracy but allowed for real-time application. Lastly, we focused on macro-averaged metrics to better reflect performance across underrepresented groups, and we removed rare or noisy samples to enhance model robustness. These decisions reflect a deliberate balance between predictive power, fairness, and operational feasibility.

11. Ethical Challenges and Considerations

Although the project did not involve high-stakes decision-making or external deployment, several ethical considerations emerged during its development, primarily in the domains of **data privacy**, **fairness**, and **model transparency**.

Data Privacy:

During the preprocessing and exploration phases, we identified that some of the datasets—particularly free-text fields and comments—contained Personally Identifiable Information (PII), such as employee names or references to specific team members. While these were not central to the modeling pipeline, their presence raised concerns about the responsible handling of sensitive internal data. As a precaution, we excluded personal or irrelevant textual content from the features used in modeling. Additionally, we collaborated with Nokia to identify which fields could be retained, anonymized, or removed altogether prior to training.

Fairness and Bias:

An analysis of the model's performance across different support groups



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revealed noticeable disparities. While groups such as **SS7**, **UI**, and **Doc** achieved high scores across all metrics (above 0.8 F1), other groups—most notably **SPV**, **Support**, and **VZ Support**—had significantly lower performance (F1 scores ranging from 0.2 to 0.45). These gaps suggest potential bias caused by data imbalance, underrepresentation, or similarity between classes. Although the model performs well on average, these discrepancies highlight the importance of monitoring and mitigating bias to ensure fair and consistent treatment across all groups.

Model Interpretability:

As the final model is a neural network, it behaves largely as a “black box.” This lack of interpretability presents a challenge for stakeholders and end users who may wish to understand why a particular team or group was predicted for a given ticket. While neural networks were chosen for their effectiveness in modeling complex relationships across heterogeneous data, they inherently lack transparency. In future iterations, the use of explainable AI (XAI) tools—such as SHAP values or attention layers—could help uncover the driving factors behind predictions and promote trust in the system’s outputs.

Conclusion:

In summary, while the ethical risks encountered in this project were relatively limited, they underscore the need for caution when handling internal organizational data, and for continuous evaluation of fairness and interpretability in machine learning systems.

12. Summary

This project demonstrated the feasibility and impact of applying machine learning to optimize fault resolution workflows within a complex organizational setting such as Nokia. By integrating structured metadata with unstructured ticket text using embeddings, we



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developed a robust classification model to support automated ticket triage. This solution aims to reduce manual effort, improve response times, and enable smarter resource allocation.

A pivotal point in the project was the refinement of the research question. Initially, the focus was on predicting specific obstacles in the fault resolution process, such as delays or handovers. However, after consultations with Nokia stakeholders, we reframed the question toward a broader and more business-aligned goal: *Can machine learning optimize issue resolution times in telecommunications?* This shift guided both modeling strategies and evaluation metrics, aligning technical outcomes with operational impact.

The model achieved strong performance indicators, including an F1 score above 0.6 and an AUC-ROC approaching 0.9, reflecting reliable discrimination between support groups. Group-level classification outperformed team-level prediction, primarily by addressing label sparsity and boosting generalization. Iterative experimentation, facilitated by tools like Weights & Biases, enabled effective hyperparameter tuning and training oversight.

The model's learning behavior was consistent, with high accuracy, strong F1 scores, and stable convergence. While it performed well across dominant groups such as CNF, UI, and SS7, it faced challenges in distinguishing overlapping or underrepresented classes like Support and SPV. Still, with a test accuracy of 61.8%, the model proved reliable and provides a solid foundation for further improvements, such as targeted training or data augmentation.

Importantly, these results validated the redefined research question by enabling accurate group-level predictions at ticket creation, reducing misassignments and delays, and contributing to operational efficiency.

Beyond predictive accuracy, the project also emphasized ethical data handling, particularly due to the sensitivity of internal organizational data. Performance disparities across groups highlighted the importance of fairness monitoring and suggested avenues for improvement through data balancing techniques.

In conclusion, the project successfully delivered a scalable and functional ML solution for ticket triage at Nokia, laying the groundwork for broader applications in intelligent fault assignment and service



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process optimization. Future directions include improving model interpretability, enabling real-time deployment, and expanding into multitask learning with feedback-driven refinement.

13. References

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<https://arxiv.org/pdf/2207.11187>
2. *Using Machine Learning for Automatic Bug Triage*
<https://arxiv.org/abs/2307.00009>
3. *Intelligent Ticket Assignment System Using Machine Learning and Rule Engines*
<https://arxiv.org/abs/1808.02636>

14. Appendices

Figures 1:EDA's graphs

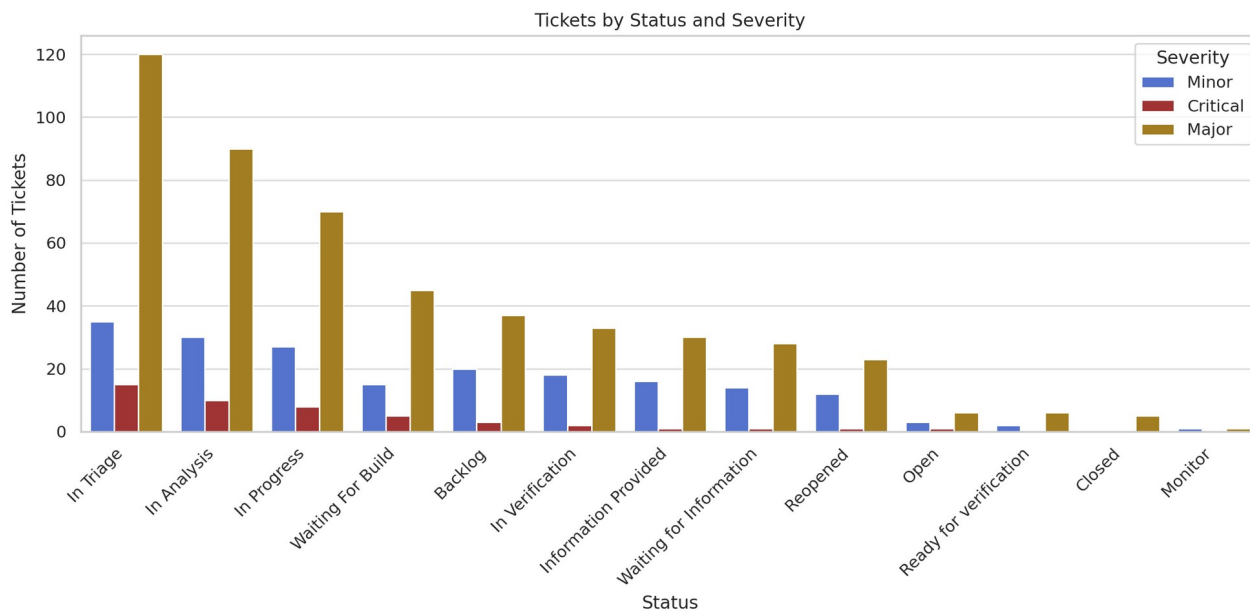


Fig 1.1: Tickets by Status and Severity

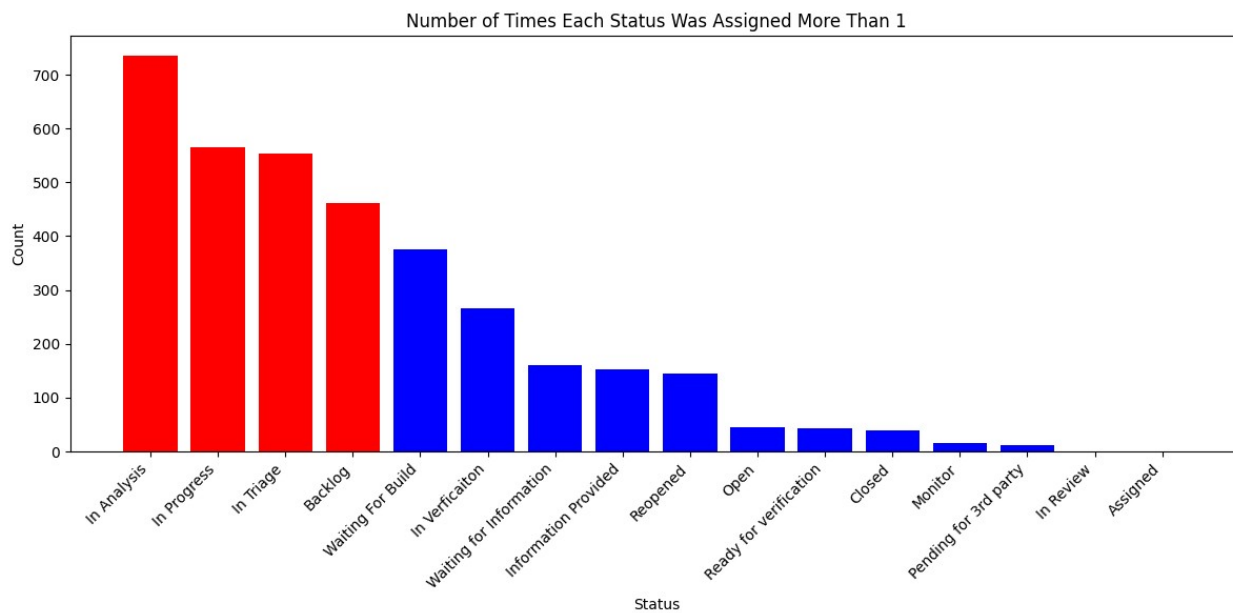


Fig 1.2: Number of Times Each Status Was Assigned More Than 1

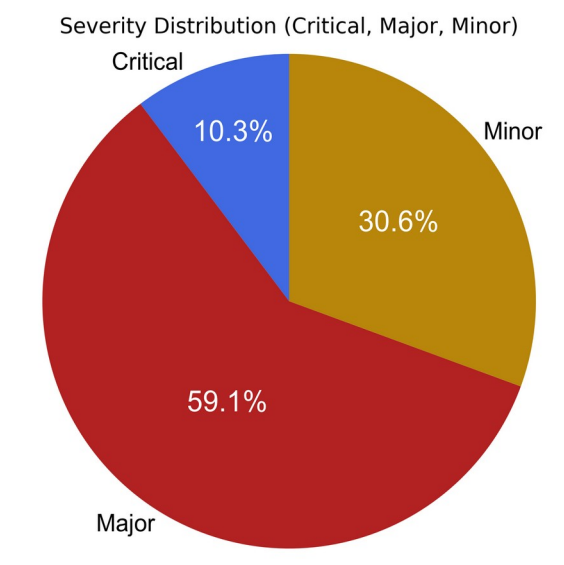


Fig 1.3:Severity Distribution

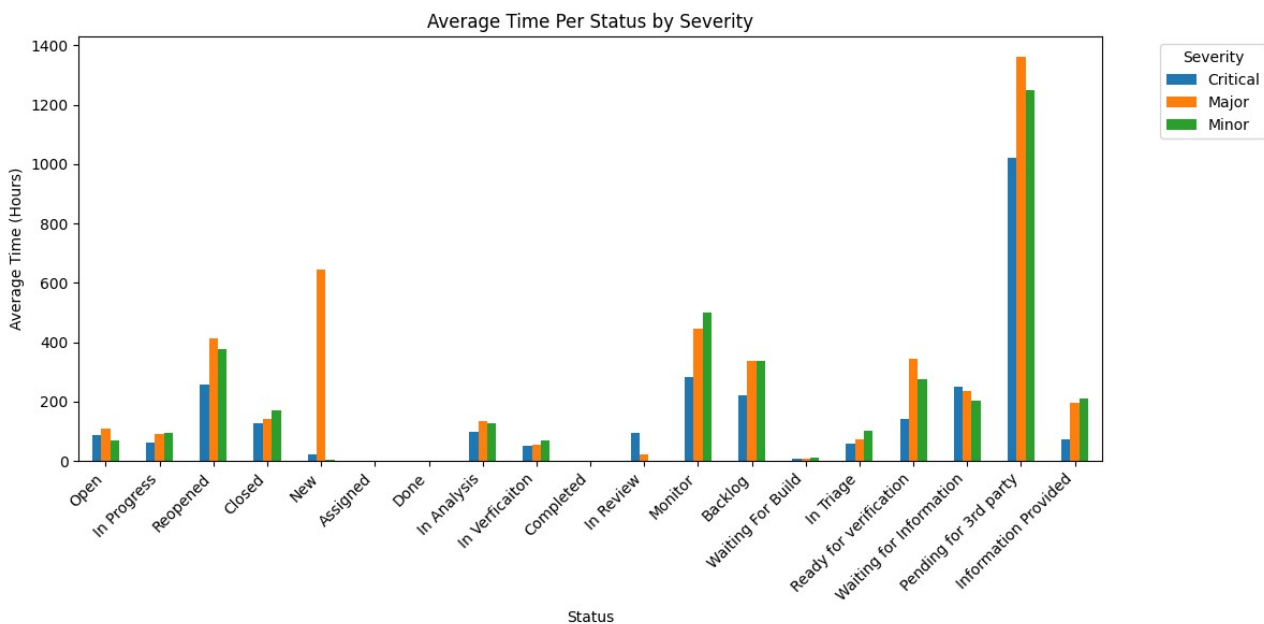


Fig 1.4:Average Time Per Status by Severity



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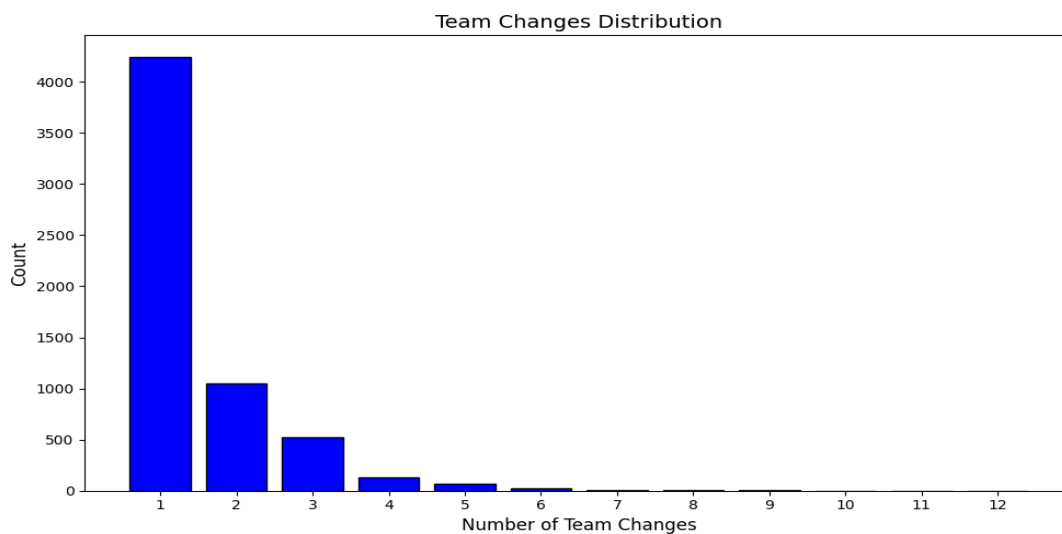


Fig 1.5:Team Changes Distribution

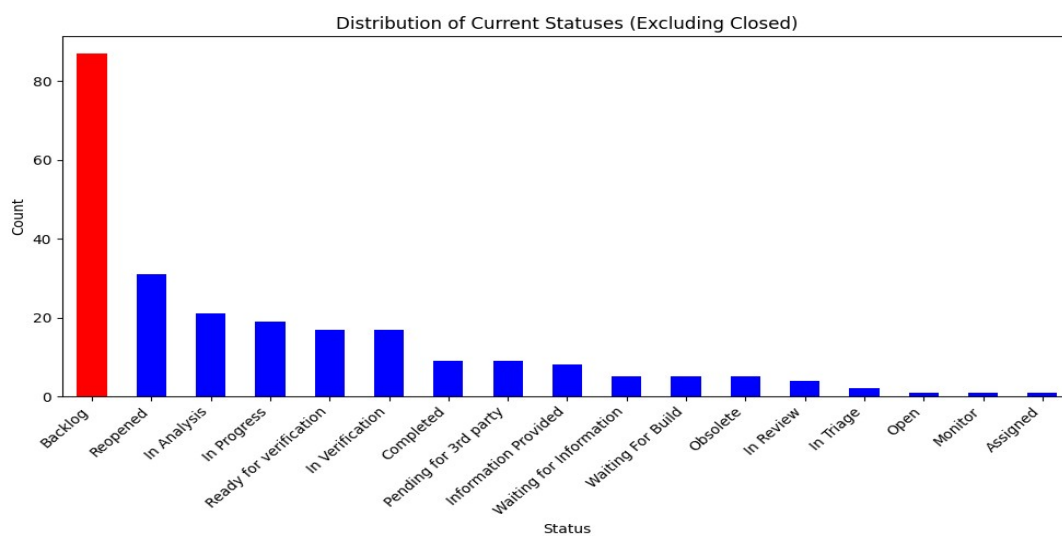


Fig 1.6:Distribution of Current Statuses (Excluding Closed)



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Figure 2:Hyperparameter tuning

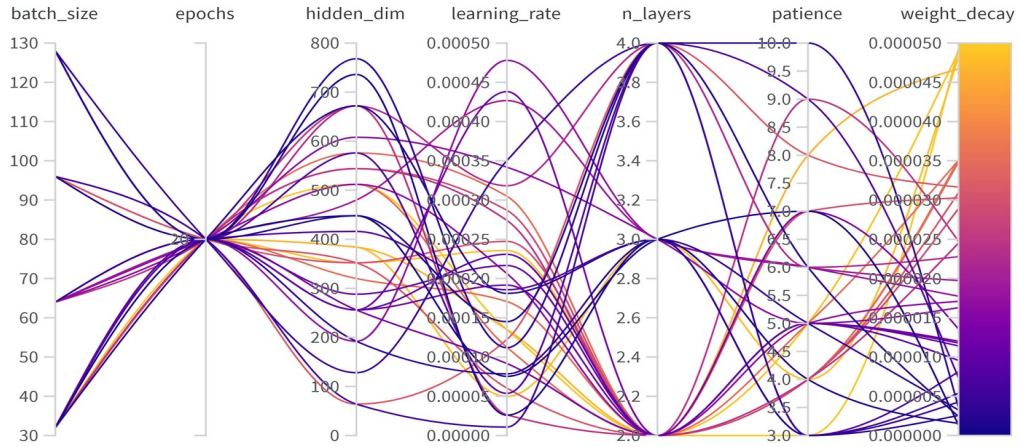


Fig. 2.1:Hyperparameter tuning of the model

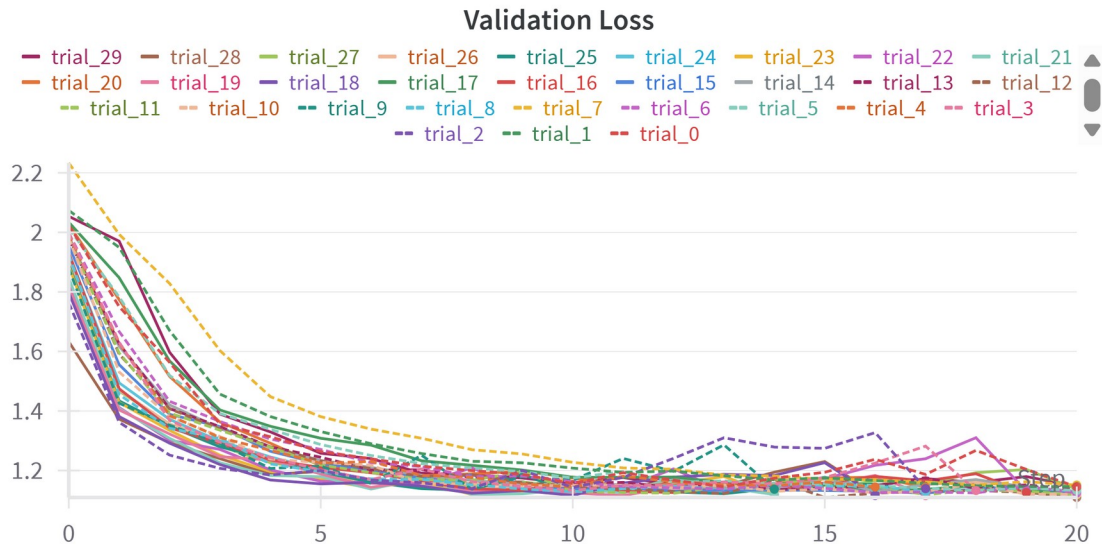


Fig. 2.2:Validation Loss



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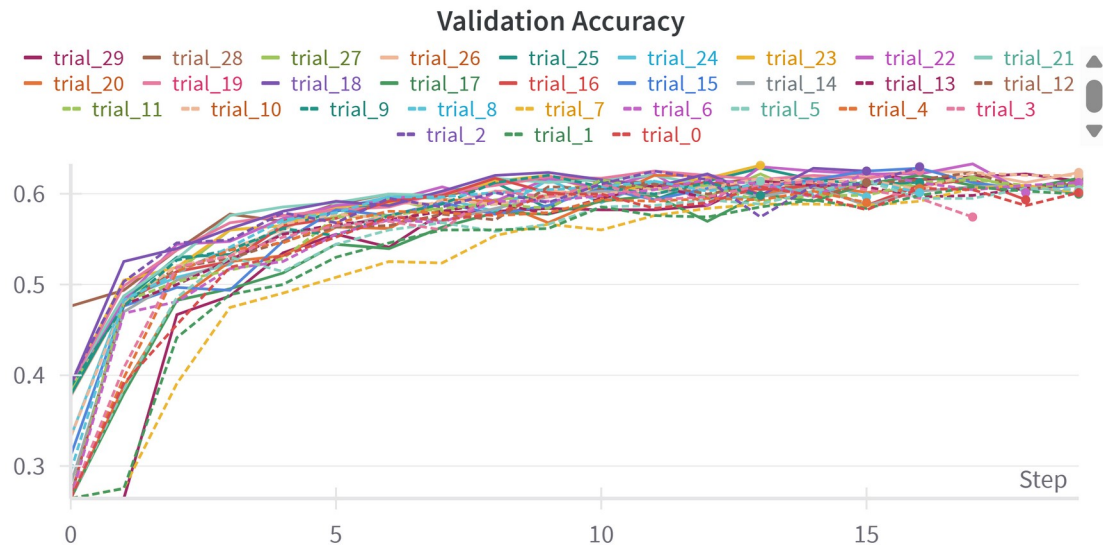


Fig. 2.3: Validation Accuracy

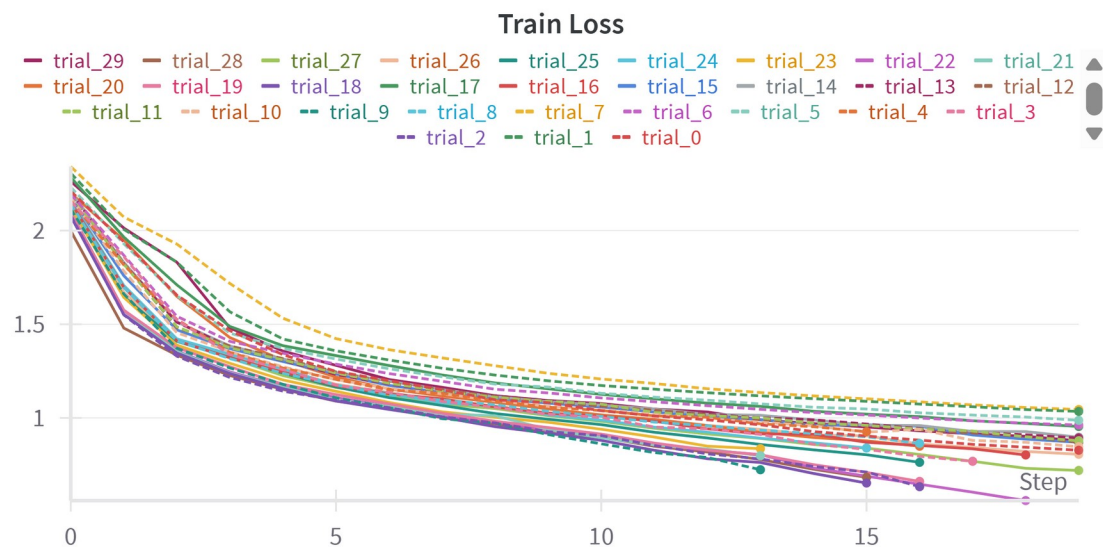


Fig. 2.4: Train Loss



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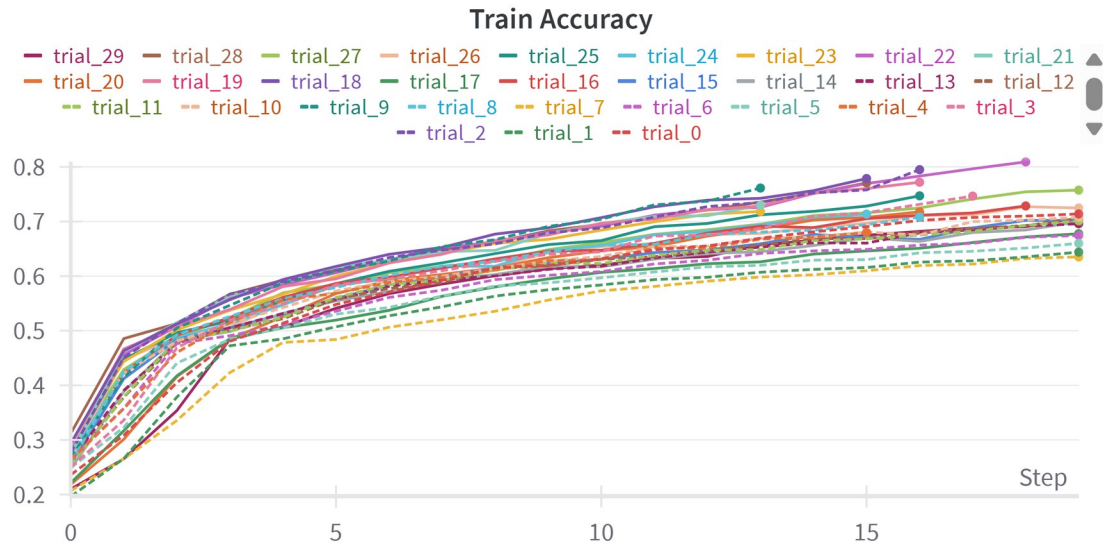


Fig. 2.5: Train Accuracy

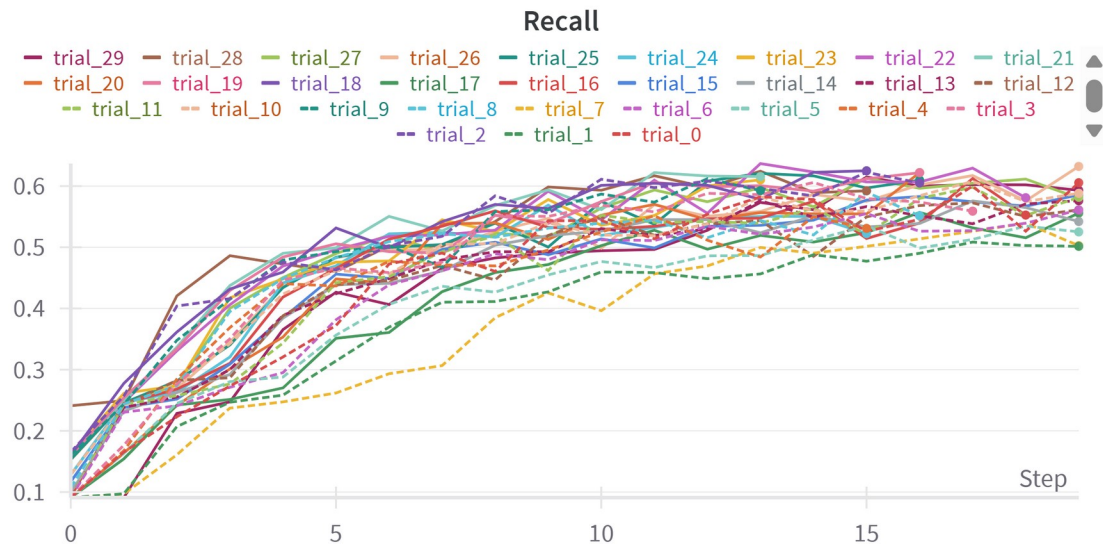


Fig. 2.6: Recall



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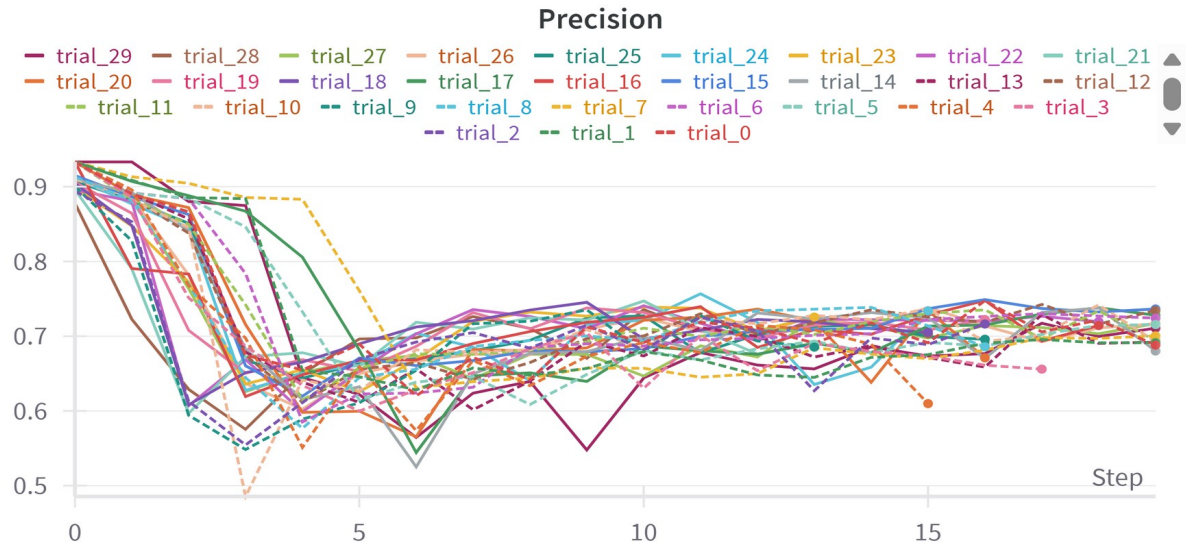


Fig. 2.7: Precision

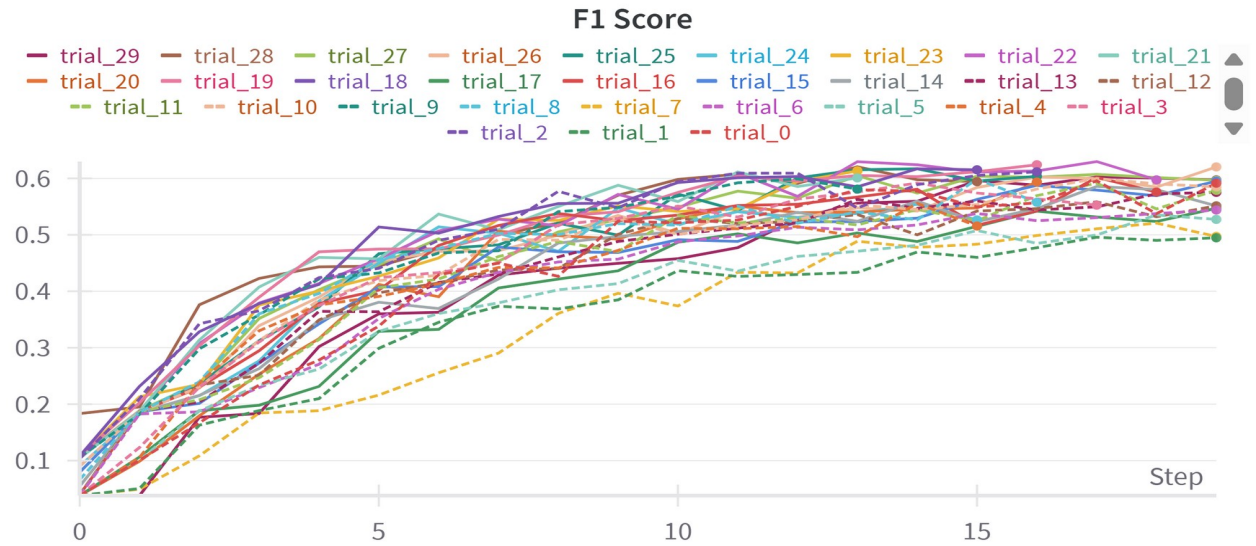


Fig. 2.8: F1 Score



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אוניברסיטת תל אביב



Figure 3: Model selection and hyperparameters optimization

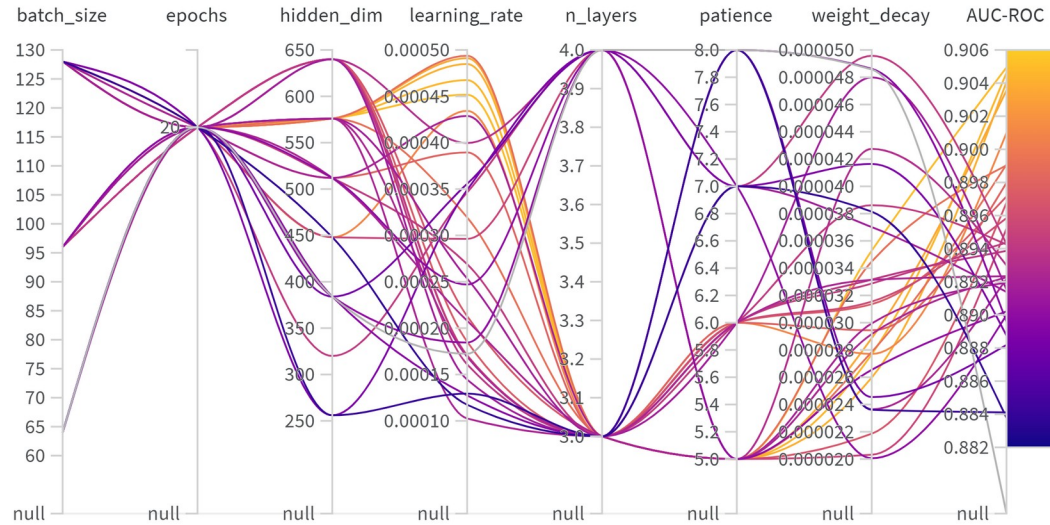


Figure 4: Model Results and Performance

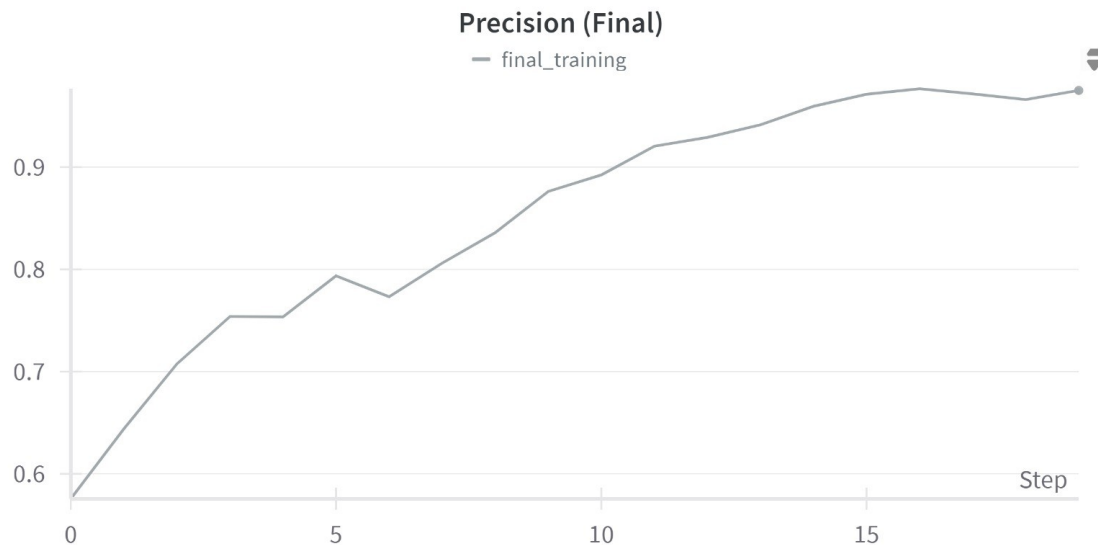


Fig. 4.1: Precision of Final model

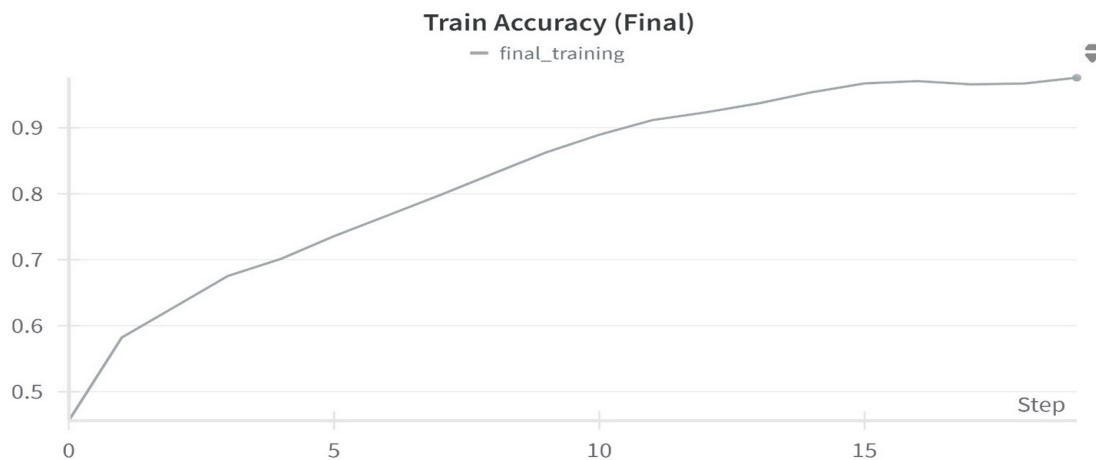


Fig. 4.2: Train Accuracy of Final model

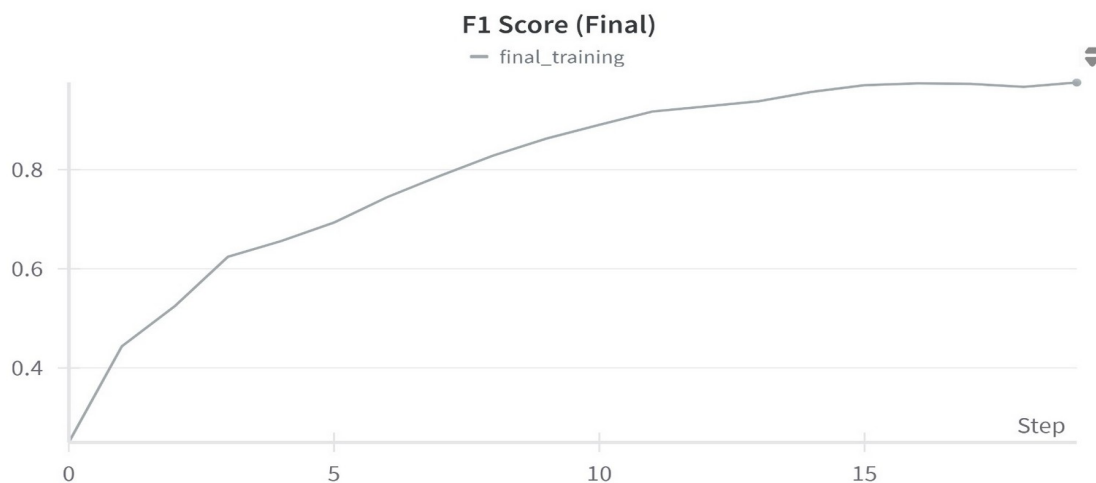


Fig. 4.3: F1 Score of Final model

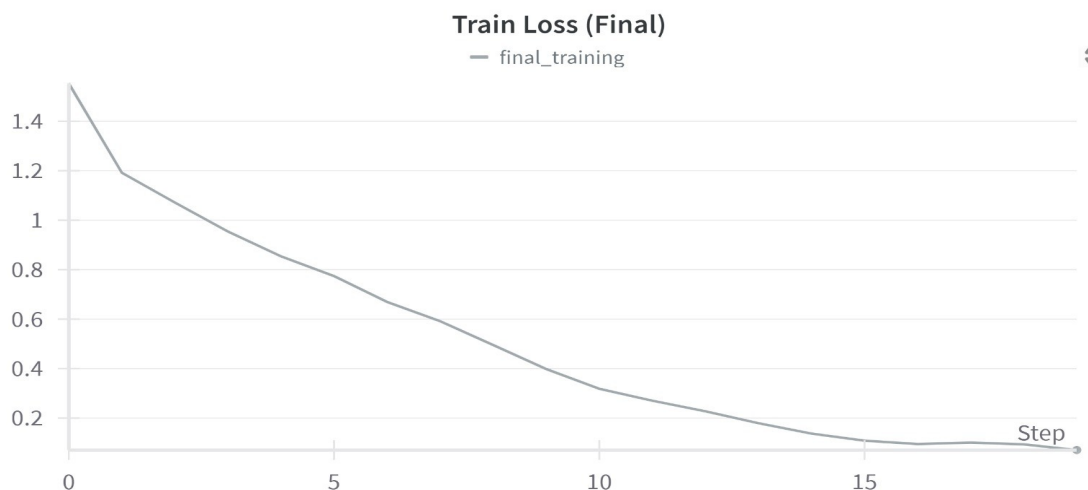
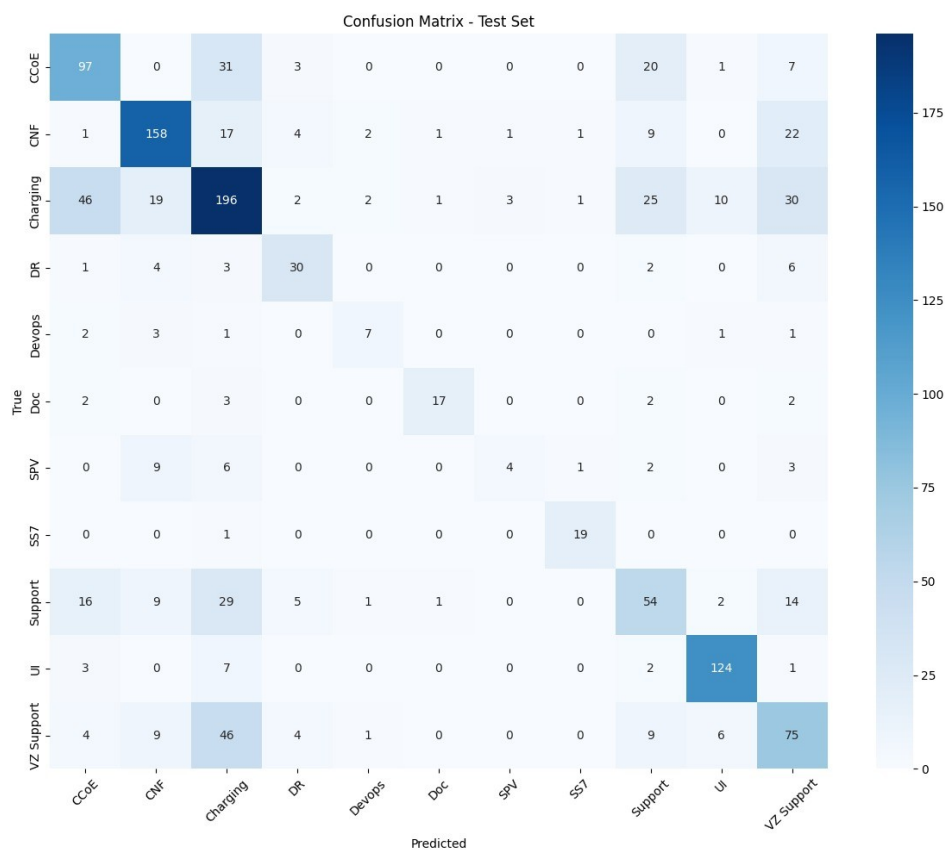


Fig. 4.4: Train Loss of Final model

Figure 5: Test set confusion matrix





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Figure 6: metrics per group scores

